

Section 7 RAG-QA Service Technical Constraints

7.1 Non-Functional Baseline Requirements

1. End-to-end latency: 90% percent of user queries must complete within 10 seconds end-to-end (HTTP request received to full response returned to client).
2. Supported corpus input formats: native PDF, DOCX, plain text, scanned-image-based PDF (requires dedicated OCR pre-processing pipeline).
3. Corpus language: mixed bilingual Chinese-English content. System shall handle code-switching within single document chunks.
4. Retrieval baseline configuration: default top_k = 4 candidate chunks; reranker module toggle-able ON / OFF; generation temperature configurable within range 0.1 ~ 0.9.
5. Maximum input context window for LLM calls shall not exceed model context-window limit; implement truncation strategy when retrieved context overflows.

90%请求端到端响应时间 ≤ 10 秒；语料支持原生PDF、DOCX、文本、扫描PDF；支持中英混合文档；top_k默认4；重排器可开关；temperature取值0.1-0.9；检索上下文溢出时执行截断策略。

7.2 Error Handling Specification

- Low-similarity retrieval condition: similarity score below threshold = 0.45. System returns explicit refusal message, must not hallucinate plausible-sounding answers.
- Zero retrieved chunks: return capability boundary notice to end-user.
- Multi-turn dialogue: conversation context state maintained per session-ID. Users do not need to repeat historical question context for follow-up questions. Session idle timeout set to 30 minutes. After timeout session context is discarded.
- API error codes: HTTP 400 for bad user input; HTTP 429 for rate-limiting; HTTP 500 for internal service failure.

相似度低于0.45阈值拒绝回答；无检索结果告知能力边界；多轮会话依靠session-id保存上下文，空闲30分钟销毁会话；定义API错误码：400错误输入，429限流，500内部异常。

7.3 Observability & Logging Mandatory Fields

Every single user request must log:

- session_id, timestamp, raw user query string
- list of retrieved chunks: source filename, file_type(native-pdf / scanned-pdf), page number, chunk

snippet, similarity score

- final generated answer text
- token_in (prompt tokens), token_out (completion tokens)
- end-to-end latency in milliseconds
- runtime parameters: top_k value, reranker enabled flag, temperature setting
- detection flags: prompt-injection-detected flag, pii_detected flag

所有请求日志字段包含会话ID、时间戳、用户query、检索片段元数据、输入输出token、延迟毫秒、运行参数、提示注入标记、PII检测标记。

7.4 Cost Estimation & Sensitivity Analysis Requirement

Deliver token-cost estimation per 1000 API invocation.

Sensitivity analysis must cover three variable dimensions:

1. top_k variations (e.g. 2,4,6)
2. reranker mode: ON versus OFF
3. temperature values (e.g. 0.1,0.4,0.7)

Report how token consumption, latency and expected answer quality shift across different configurations.

需要输出每1000次API调用token成本估计。敏感性分析覆盖top_k(2,4,6)、reranker开关、temperature(0.1,0.4,0.7)，分析不同配置下token消耗、延迟、回答质量变化。

7.5 Evolvability Requirements

RAG components shall be loosely coupled: embedding model, chunking logic, reranker module, LLM generator should be swap-able without large-scale code refactoring.

New knowledge documents shall be ingested without full vector-database rebuild.

系统组件解耦：Embedding、分块、重排器、生成模型可替换；新增知识库文档不需要重建全部向量库。